THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION MIL-N-25027

DD Form 672, MAY 88 PREVIOUS EDITIONS ARE OBSOLETE

PREPARING ACTIVITY DLA-IS

CUSTODIANS ARMY- AV NAVY- AS AIR FORCE- 11 DLA-

REVIEW 82
USER CR MI
PROJECT NUMBER 5310-1955

DISTRIBUTION STATEMENT A. Approved for public release distribution is unlimited

MILITARY SPECIFICATION SHEET

TITLE NUT, SELF-LOCKING, PLATE, TWO-LUG, REDUCED RIVET SPACING, LOW HEIGHT, STEEL, 125 KSI Ftu, 450°F

AMSC- N/A

SUPERSEDING MS21069 F 30 SEPT 87 (SEE NOTE 5)

FSC 5310

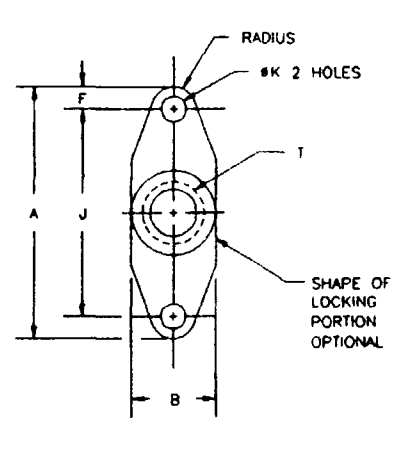
SPECIFICATION SHEET NUMBER MS21069 1 FEB 94 REV G

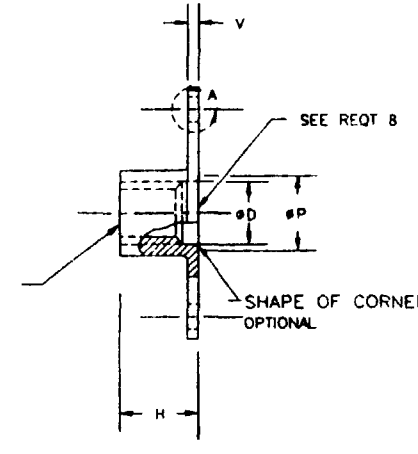
Page 1 of 2

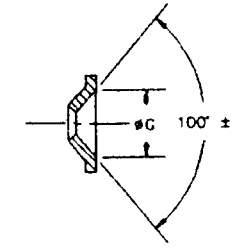
INCH-POUND

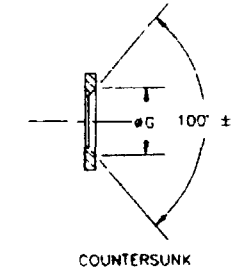
Ⓖ DENOTES CHANGE(S)

CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS AS SPECIFIED HEREIN









VIEW A

TABLE I DASH NUMBERS AND DIMENSIONS

DASH NUMBERS				T THREAD SIZES	A	B		ØD	F	ØG	H 1/	J 2/	ØK 3/ +.005 −.000	ØP	V	AXIAL STRENGTH LBS/MIN	WEIGHT LBS/100
PLAIN RIVET HOLES		COUNTERSUNK OR DIMPLED RIVET HOLES															
NON-DRY FILM LUBRICANT	DRY FILM LUBRICANT	NON-DRY FILM LUBRICANT	DRY FILM LUBRICANT		MAX	MAX	MIN	MIN	MIN	±.010	MAX	±.002		MAX	MAX		
-02	L02	–	–	.086-56UNJC-3B	.410	.162	.140	–	.071	4/	.110	.250	.066	.130	.035	440	05
-03	L03	–	–	.099-48UNJC-3B	.425	.184	.140	–	.071	4/	.125	.265	.066	.143	.035	550	08
-04	L04	-04K	L04K	.112-40UNJC-3B	.630	.260	.140	–	.089	.200	.143	.408	.098	.166	.040	750	11
-06	L06	-06K	L06K	.138-32UNJC-3B	.661	.265	.195	–	.089	.200	.171	.437	.098	.206	.047	1,130	14
-08	L08	-08K	L08K	.164-32UNJC-3B	.692	.297	.230	.168	.100	.200	.250	.468	.098	.248	.047	1,720	18
-3	L3	-3K	L3K	.190-32UNJF-3B	.724	.328	.298	.194	.100	.200	.250	.500	.098	.274	.047	2 460	26
-4	L4	-4K	L4K	.250-28UNJF-3B	.786	.414	.380	.254	.100	.200	.281	.562	.098	.344	.055	4 580	40
-5	L5	-5K	L5K	.3125-24UNJF-3B	1.006	.505	.475	.317	.125	.230	.328	.718	.130	.417	.065	7 390	77
-6	L6	–	–	.375-24UNJF-3B	1.116	.614	.580	.379	.125	.230	.344	.828	.130	.505	.075	11,450	1 86

1/ MINIMUM "H" NOT SPECIFIED LIMITED ONLY BY STRENGTH REQUIREMENTS OF SPECIFICATION
2/ CENTER OF THE TAPPED HOLE SHALL NOT DEVIATE IN ANY DIRECTION FROM THE CENTER OF THE PLATE NUT AS DETERMINED BY THE RIVET HOLES BY MORE THAN .005
3/ DIMPLED RIVET HOLE TOLERANCE FOR "ØK" IS +.015 −.000
4/ SPACE WILL NOT PERMIT COUNTERSINK

Form Approved OMB No 0704-0188

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:26Z
Check the source to verify that this is the current version before use.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION MIL-N-25027

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

Form Approved
OMB No 0704-0188

REQUIREMENTS

(G) 1 MATERIAL CARBON STEEL COMPOSITIONS 1035 (UNS G10350) 1040 (UNS G10400) AND 1050 (UNS G10500) IN ACCORDANCE WITH ASTM A827 OR QQ-S-700 1042 (UNS G10420) IN ACCORDANCE WITH ASTM A29 ALLOY STEEL GRADES 4130 (UNS G41300) 4140 (UNS G41400) AND 8740 (UNS G87400) IN ACCORDANCE WITH ASTM A29

2 FINISH CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416 TYPE II CLASS 3 FOR DRY FILM LUBRICATED NUTS THE TYPE AND CLASS ARE OPTIONAL IF THE NUTS MEET THE SALT SPRAY REQUIREMENTS OF QQ-P-416 TYPE II

3 DIMENSIONING AND TOLERANCING DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14.5M

4 HARDNESS 49 HRC MAX

5 THREADS THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-S-8879

6 SURFACE TEXTURE SURFACE TEXTURE UNLESS OTHERWISE SPECIFIED SHALL NOT EXCEED 125 MICROINCHES IN ACCORDANCE WITH ANSI/ASME B46.1

7 LUBRICANT DRY FILM LUBRICANT APPROVED IN ACCORDANCE WITH MIL-N-25027 NON-DRY FILM LUBRICANTS SHALL BE SOLUBLE IN THE CLEANER SPECIFIED IN MIL-S-8802

(G) 8 EDGES BREAK ALL SHARP EDGES AND REMOVE ALL BURRS

9 COUNTERBORE/COUNTERSINK ON SIZE 164 AND LARGER THREAD RELIEF SHALL BE .062 MINIMUM ON SIZE 138 AND SMALLER COUNTERSINK TO "P" DIAMETER

10 PART NUMBER THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I

EXAMPLE MS21069L4K
- DASH NUMBER
- BASIC MS NUMBER

MS21069L4K INDICATES NUT SELF-LOCKING PLATE TWO LUG REDUCED RIVET SPACING LOW HEIGHT STEEL 125 KSI Ftu 450°F 250-28UNJF-3B, DRY FILM LUBRICATION COUNTERSUNK OR DIMPLED RIVET HOLES

NOTES

1 ALL DIMENSIONS ARE IN INCHES

2 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN THE TEXT OF STANDARD SHALL TAKE PRECEDENCE

3 REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN

4 DESIGN AND USAGE LIMITATIONS THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588 ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED

5 MS21069 SUPERSEDES NAS697

INTERCHANGEABILITY RELATIONSHIP

MS21069 NUTS CAN UNIVERSALLY REPLACE NAS697 NUTS OF LIKE MATERIAL THREAD SIZE LUBRICANT (DRY FILM OR NON-DRY FILM LUBRICANT) AND FASTENING METHOD (PLAIN RIVET HOLES DIMPLED OR COUNTERSUNK RIVET HOLES) BUT THESE NAS697 NUTS CANNOT UNIVERSALLY REPLACE MS21069 NUTS

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS	CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS
NAS697X02	MS21069-02	NAS697X3K	MS21069-3K
NAS697A02	MS21069L02	NAS697A3K	MS21069L3K
NAS697X03	MS21069-03	NAS697X4	MS21069-4
NAS697A03	MS21069L03	NAS697A4	MS21069L4
NAS697X04L	MS21069-04	NAS697X4K	MS21069-4K
NAS697A04L	MS21069L04	NAS697A4K	MS21069L4K
NAS697X04LK	MS21069-04K	NAS697X5	MS21069-6
NAS697A04LK	MS21069L04K	NAS697A5	MS21069L6
NAS697X06L	MS21069-06	NAS697X5K	MS21069-6K
NAS697A06L	MS21069L06	NAS697A5K	MS21069L6K
NAS697X06LK	MS21069-06K	NAS697X6	MS21069-6
NAS697A06LK	MS21069L06K	NAS697A6	MS21069L6
NAS697X08	MS21069-08		
NAS697A08	MS21069L08		
NAS697X08K	MS21069-08K		
NAS697A08K	MS21069L08K		
NAS697X3	MS21069-3		
NAS697A3	MS21069L3		

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY- AV NAVY- AS
AIR FORCE- 11 DLA-
REVIEW 82
USER CR, MI
PROJECT NUMBER 5310-1955

MILITARY SPECIFICATION SHEET

TITLE

NUT, SELF-LOCKING, PLATE TWO LUG, REDUCED RIVET SPACING LOW HEIGHT STEEL 125 KSI Ftu, 450°F

SPECIFICATION SHEET NUMBER
MS21069 1 FEB 94 REV G

SUPERSEDING
MS21069 F 30 SEPT 87 (SEE NOTE 5)

AMSC- N/A FSC 5310

DISTRIBUTION STATEMENT
A. Approved for public release distribution is unlimited

DD Form 672, MAY 88 PREVIOUS EDITIONS ARE OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:26Z
Check the source to verify that this is the current version before use.